

# Lukas Jander

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## Education

**Rochester Institute of Technology**, GPA: 3.99/4.00, Dean's List, Honors Program Rochester, NY  
BS/ME Mechanical Engineering Expected Graduation May, 2027

## Experience

**Laboratory for Laser Energetics Engineering Intern** January, 2025 – July 2026

- Designed and analyzed a flexure-based 3 degree of freedom mirror mount
  - Used Monte Carlo simulation (ANSYS + Sigfit) to quantify sensitivity of optical performance to mounting point tolerances
  - Verified diffraction limited performance at 6THz via raytracing on deformed optical surfaces
  - Ran ANSYS parametric studies to select materials and dimensions to minimize surface RMS
- Redesigned a vacuum compatible lens assembly to accommodate two optic sizes (6" and 150mm)
  - Eliminated metal on optic contact using moveable Delrin pins
  - Performed tolerance stack up to verify positioning accuracy
- Diagnosed a temperature gradient issue in an existing LN<sub>2</sub> driven argon gas cooling system
  - Measured temperature profiles to estimate convective heat transfer coefficients
  - ANSYS thermal simulations compared favorably to experimental data within 3K
- Iterated a rigging fixture design for a 60lb X-ray camera following an equipment damaging incident
  - Diagnosed cause of failure and evaluated multiple potential solutions
  - Validated redesign with ANSYS mechanical and bolted joint hand calculations
  - New design is used on multiple other cameras after successfully being deployed on X-ray camera
- Built a passive lift mechanism to reduce the perceived lift weight of a 35lb calorimeter to ~5lb
  - Ensured the design's safety and ease of use based on physical prototyping

**Insulet Corporation Quality Engineering Intern** January, 2026 – April 2026

- Wrote test procedures for cyclic load testing of a medical device
  - Programmed Instron tensile test methods to automate testing
- Created torsional test setups to evaluate repeatability of manufacturing procedures
- Characterized repeatability of surface roughness measurements to investigate ink adhesion issues

**SEPAC Inc. Engineering Intern** May, 2024 – August, 2024

- Designed a test setup to automate brake performance evaluation
  - Integrated a torque supplying motor with transducer feedback and PLC control
- Wrote programmable logic controller (PLC) code to automate test procedures
- Performed hardware and software repairs on existing test setups to minimize downtime

**Rochester Institute of Technology Researcher in a Microelectronics Lab** May, 2023 – August, 2023

- Characterized sensitivity of five different designs of a MEMS viscosity sensor to monitor oil health
- Created a high pass filtering and amplification circuit to remove DC offset and low frequency noise
- Isolated and subtracted out temperature dependent sensor drift
- Findings presented in *Waterproofing a Thermally-Actuated Vibrational MEMS Viscosity Sensor*

## Projects and Clubs

**RIT Competitive Rocketry Team: Airbrakes Subteam Member**

- Performed FEA to evaluate mechanical load cases experienced during flight
- Designed an active apogee control mechanism to meet target apogee
  - Wrote MATLAB scripts to predict load on components and optimize design
- 2025-26 rocket successfully flew within 200ft of target apogee of 10,000ft using this mechanism
- Performing computational fluid dynamics on a computing cluster to obtain drag estimates for the 2026-27 rocket

**Combination Lock Solver**

- Designed and built a device that opens combination locks and display the solution
- Arduino controlled stepper motor, solenoid, and a beam break sensor drove the mechanism
- Testing of servo motor and design of PCB in progress

**Treasurer of Rock Climbing Club**

- Organized events, fundraised, and communicated with university staff about budget

## Skills

**Software:** ANSYS, MATLAB, SigFit, LabVIEW, NX, SOLIDWORKS, Creo, Autodesk Inventor/Fusion

**Miscellaneous:** GD&T proficient, Fluent in German and Czech